

Natural Language Processing

SITAMS CSE (AI & ML) • R23 • III Year • VI Semester • Course Code: 23CAI353T

Source basis: Official SITAMS R23 CSE (AI & ML) syllabus uploaded for this Study Hub. This concise revision guide preserves the syllabus topic statements and adds a study checklist; it is not a replacement for prescribed textbooks or faculty notes.

Unit 1: INTRODUCTION TO NLP

Syllabus topics

- Introduction to NLP: Origins and Challenges, Language and Grammar in NLP, Regular Expressions and Finite-State Automata, Tokenization: Text Segmentation and Sentence Splitting, Morphological Parsing: Stemming and Lemmatization, Spelling Error Detection and Correction, Minimum Edit Distance and Applications, Statistical Language Models: Unigram, Bigram, and Trigram Models, Processing Indian Languages in NLP

Revision checklist

- Know the definitions and principles named in the syllabus.
- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.
- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

Source page(s): 180

Unit 2: WORD-LEVEL AND SYNTACTIC ANALYSIS

Syllabus topics

- Introduction, Part-of-Speech (POS) Tagging: Rule-Based, Stochastic and Transformation- Based Approaches, Hidden Markov Models (HMM) and Maximum Entropy Models for POS Tagging, Context-Free Grammar (CFG) and Constituency Parsing, Treebanks and Normal Forms for Grammar, Top-Down and Bottom-Up Parsing Strategies, CYK Parsing Algorithm, Probabilistic Context-Free Grammars (PCFGs), Feature Structures and Unification

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Unit 3: TEXT CLASSIFICATION AND INFORMATION RETRIEVAL

Syllabus topics

- Naïve Bayes Classifier for Text Classification, Training and Optimization for Sentiment Analysis, Information Retrieval: Basic Concepts and Design Features, Information Retrieval Models: Classical, Non-Classical, and Alternative Models, Cluster Model, Fuzzy Model, and LSTM-Based Information, Retrieval, Word Sense Disambiguation (WSD) Methods: Supervised and Dictionary-Based Approaches

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Unit 4: MACHINE TRANSLATION AND SEMANTIC PROCESSING

Syllabus topics

- Introduction to Machine Translation (MT), Language Divergence and Typology in MT Encoder- Decoder Model for Machine Translation, Translating in Low-Resource Scenarios, MT Evaluation Metrics and Techniques, Bias and Ethical Issues in NLP and Machine Translation, Semantic Analysis and First-Order Logic in NLP, Thematic Roles and Selectional Restrictions in Semantics, Word Senses and Relations Between Senses

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Unit 5: SPEECH PROCESSING AND ADVANCED NLP MODELS

Syllabus topics

- Speech Fundamentals: Phonetics and Acoustic Phonetics, Digital Signal Processing in Speech Analysis, Feature Extraction in Speech: Short-Time Fourier Transform (STFT), Mel- Frequency Cepstral Coefficients (MFCC) and Perceptual Linear Prediction (PLP), Hidden Markov Models (HMMs) in Speech Recognition

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Study Hub status

This guide is syllabus-derived. External resources and quiz questions are tracked separately and are only labelled verified after source checking.